NOTES ON SHEAVES AND DESCENT

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Abstract. We collect some basic facts on sheaves on sites, descent for quasi-coherent modules, and the Yoneda lemma. These notes serve as a test document.

Contents

1. Introduction 1
2. The Yoneda lemma 1
3. Descent 2
4. Sheaves on sites 2
4.1. Long computation 3
4.2. Filler subsection 1 3
4.3. Filler subsection 2 3
4.4. Filler subsection 3 4
4.5. Filler subsection 4 4
4.6. Filler subsection 5 4
4.7. Filler subsection 6 5
References 5

1. Introduction

Let X be a scheme and $\mathcal{O}_X$ its structure sheaf. The theory of descent, initiated by Grothendieck [Gro71], is treated at length in [Vis05]. We follow the conventions of [The26]. In Section 2 we recall the Yoneda lemma (Lemma 2.1), and in Section 3 we prove Theorem 3.2.

2. The Yoneda lemma

Let $\mathcal{C}$ be a locally small category and write $h_X = \mathrm{Hom}_{\mathcal{C}}(-, X)$.

Lemma 2.1 (Yoneda)**.** *For every presheaf $F\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ and every object X of $\mathcal{C}$ there is a bijection, natural in X and F,*

$$\mathrm{Hom}_{\mathrm{PSh}(\mathcal{C})}(h_X, F) \xrightarrow{\sim} F(X), \qquad \eta \longmapsto \eta_X(\mathrm{id}_X).$$

Proof. Given $a \in F(X)$ define $\eta_Y^a(g) = F(g)(a)$ for $g\colon Y \to X$. Naturality of η forces the commutative square

$$\begin{array}{ccc} h_X(X) & \xrightarrow{\eta_X} & F(X) \\ {\scriptstyle g^*}\downarrow & & \downarrow{\scriptstyle F(g)} \\ h_X(Y) & \xrightarrow[\eta_Y]{} & F(Y) \end{array}$$

and chasing id_X gives $\eta_Y(g) = F(g)(\eta_X(\mathrm{id}_X))$. □

Corollary 2.2. *The Yoneda embedding $h\colon \mathcal{C} \to \mathrm{PSh}(\mathcal{C})$ is fully faithful.*

This is used throughout; see [Mac98].

3. Descent

Definition 3.1. Let $f\colon S' \to S$ be a morphism of schemes. A *descent datum* for a quasi-coherent $\mathcal{O}_{S'}$-module $\mathcal{F}'$ is an isomorphism $\varphi\colon p_1^*\mathcal{F}' \xrightarrow{\sim} p_2^*\mathcal{F}'$ on $S'' = S' \times_S S'$ satisfying the cocycle condition

$$p_{13}^*\varphi = p_{23}^*\varphi \circ p_{12}^*\varphi \quad \text{on } S' \times_S S' \times_S S'. \tag{1}$$

Theorem 3.2 (fpqc descent)**.** *Let $f\colon S' \to S$ be faithfully flat and quasi-compact. Then the functor*

$$\mathrm{QCoh}(S) \longrightarrow \mathrm{QCoh}(S' \to S), \qquad \mathcal{F} \longmapsto (f^*\mathcal{F}, \mathrm{can})$$

is an equivalence of categories.

Proof. By [The26, Tag 023N] it suffices to treat the affine case $S = \operatorname{Spec} A$, $S' = \operatorname{Spec} B$ with $A \to B$ faithfully flat. Consider the Amitsur complex

$$0 \to M \to M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B \to \cdots \tag{2}$$

which is exact since it becomes split exact after the faithfully flat base change $- \otimes_A B$. The cocycle condition (1) is exactly what is needed to descend $\mathcal{F}'$. □

Remark 3.3. The diagram

$$S' \times_S S' \underset{p_2}{\overset{p_1}{\rightrightarrows}} S' \xrightarrow{\ f\ } S$$

is a coequalizer in the category of schemes when f is fpqc.

4. Sheaves on sites

Let $\mathcal{C}$ be a site. A presheaf $\mathcal{F}$ on $\mathcal{C}$ is a sheaf if for every covering $\{U_i \to U\}_{i\in I}$ the sequence

$$\mathcal{F}(U) \to \prod_{i\in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}(U_i \times_U U_j)$$

is an equalizer. The category $\mathrm{Sh}(\mathcal{C})$ is a Grothendieck topos.

Proposition 4.1. *The inclusion* $\mathrm{Sh}(\mathcal{C}) \hookrightarrow \mathrm{PSh}(\mathcal{C})$ *admits an exact left adjoint* $\mathcal{F} \mapsto \mathcal{F}^{\#}$.

Proof. Apply the plus construction twice: $\mathcal{F}^{\#} = (\mathcal{F}^{+})^{+}$, where

$$\mathcal{F}^{+}(U) = \operatorname*{colim}_{\mathcal{U} \in J(U)} \check{H}^{0}(\mathcal{U}, \mathcal{F}).$$

Exactness follows since filtered colimits commute with finite limits. □

Example 4.2. For X a topological space, the small site $\mathrm{Open}(X)$ gives the usual notion. For a scheme X the étale site $X_{\text{ét}}$ gives étale sheaves, and

$$H^{i}_{\text{ét}}(X, \mu_n) \cong H^{i}(X(\mathbb{C}), \mathbb{Z}/n)$$

for X smooth over $\mathbb{C}$ by Artin's comparison theorem.

4.1. **Long computation.**

$$\mathrm{Hom}_{\mathrm{Sh}(\mathcal{C})}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PSh}(\mathcal{C})}(\mathcal{F}, \mathcal{G}) \tag{3}$$

$$\cong \lim_{U \in \mathcal{C}} \mathrm{Hom}_{\mathrm{Set}}(\mathcal{F}(U), \mathcal{G}(U)) \tag{4}$$

$$\cong \int_{U \in \mathcal{C}} \mathcal{G}(U)^{\mathcal{F}(U)}. \tag{5}$$

Equations (3)–(5) combine to give the end formula.

4.2. **Filler subsection 1.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. For every $n \geq 1$ we have

$$\sum_{k=0}^{n} \binom{n}{k} x^{k} y^{n-k} = (x+y)^{n}, \qquad \int_{0}^{1} t^{1}\, dt = \frac{1}{2}.$$

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4.3. **Filler subsection 2.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. For every $n \geq 2$ we have

$$\sum_{k=0}^{n} \binom{n}{k} x^{k} y^{n-k} = (x+y)^{n}, \qquad \int_{0}^{1} t^{2}\, dt = \frac{1}{3}.$$

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voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. See Theorem 3.2 and [Vis05]. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse.

4.4. **Filler subsection 3.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. For every $n \geq 3$ we have

$$\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = (x+y)^n, \qquad \int_0^1 t^3 \, dt = \frac{1}{4}.$$

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4.5. **Filler subsection 4.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. For every $n \geq 4$ we have

$$\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = (x+y)^n, \qquad \int_0^1 t^4 \, dt = \frac{1}{5}.$$

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4.6. **Filler subsection 5.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. For every $n \geq 5$ we have

$$\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = (x+y)^n, \qquad \int_0^1 t^5 \, dt = \frac{1}{6}.$$

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4.7. **Filler subsection 6.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. For every $n \geq 6$ we have

$$\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = (x+y)^n, \qquad \int_0^1 t^6 \, dt = \frac{1}{7}.$$

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